

# SwipeScreen

## Instruction Manual

SwipeScreen is a browser-based app for screening articles during a systematic review. It turns title and abstract screening into fast, swipe-style decisions, calculates inter-reviewer agreement, and exports reconciled, PRISMA-ready results.

**Everything runs in your browser.** Your search results, decisions, and exports are stored locally (IndexedDB) on your own device. Nothing is uploaded to a server, and there are no accounts to create.

## The screening pipeline

SwipeScreen is organized as a three-stage pipeline, shown as three numbered cards on the home screen:

1. **Title Screening** — screen records by title (fast, swipe-based decisions).
2. **Synthesis** — combine several reviewers' title-screening exports, measure agreement, and resolve conflicts into a single reconciled list.
3. **Abstract Screening** — screen articles by reading the full abstract.

A typical two-reviewer review runs straight down the pipeline: each reviewer does Title Screening independently, their exports are combined in Synthesis, and the reconciled list feeds Abstract Screening.

You do **not** have to use every stage. In particular, Abstract Screening can be started on its own over a complete article list (see *Abstract Screening* → *Starting from a full, un-reconciled list*).

## Getting started

### Run the app

```
npm install
npm run dev
```

Then open the local URL that Vite prints (the app is served under the `/SwipeScreen/` path). A production build is produced with `npm run build`.

### Uploading search results

Every new screening project starts by uploading a CSV of records. SwipeScreen recognizes common database exports and auto-maps their columns:

- PubMed
- Scopus
- Web of Science
- EMBASE
- Cochrane

If the format isn't recognized, columns are matched by header name, and you can always adjust the mapping by hand on the **Map Columns** step. Only **Title** is strictly required. For Abstract Screening you will also want **Abstract** mapped so each card has something to read. Files may be .csv or .txt.

Records are imported in a randomized order to reduce ordering bias during screening.

## Stage 1 — Title Screening

Create a project from the **Title Screening** → **+ New** button, enter a project name and your reviewer name/initials, upload your CSV, confirm the column mapping, and start.

Each record appears as a swipeable card showing the title and basic metadata (authors, year, journal, and links to DOI / PubMed where available).

| Action       | Swipe | Keys   | Meaning                |
|--------------|-------|--------|------------------------|
| <b>Yes</b>   | Right | → or K | Include                |
| <b>No</b>    | Left  | ← or J | Exclude                |
| <b>Maybe</b> | Up    | ↑ or M | Undecided / borderline |

Other shortcuts: Ctrl+Z undo the last decision, ? show/hide the shortcut help, Esc return home. You can also use the on-screen **No / Maybe / Yes** buttons.

Progress is saved continuously. Use **Save** on the screening screen (or **Save Progress (JSON)** on the results screen) to download a resumable snapshot, and **Resume** on the home screen to reload it later.

When every card is decided you land on the **Results** screen. Export the decisions as a CSV — this per-reviewer CSV is exactly what the Synthesis stage expects.

## Stage 2 — Synthesis

Synthesis combines the title-screening exports of two or more reviewers.

1. Open **Synthesis** → **Start Synthesis**.
2. Add each reviewer's exported CSV (at least two).
3. SwipeScreen matches records across reviewers — exactly by DOI/title, with fuzzy title matches offered for your approval — and reports agreement: **Cohen's kappa** and **PABAK** for two reviewers, **Fleiss' kappa** for three or more. Agreement statistics are computed only over records matched across all reviewers.
4. Resolve any conflicts (records the reviewers decided differently).

The output of Synthesis is a **reconciled CSV**: one row per article carrying a `final_decision` column with values such as `include`, `maybe`, or `exclude`. That reconciled file is the classic input to Abstract Screening.

## Stage 3 — Abstract Screening

Abstract Screening is where you read each abstract in full and make an include / maybe / exclude decision. Create a project from **Abstract Screening** → **+ New**. In addition to the project name and reviewer, you can

enter:

- **Inclusion criteria** and **Exclusion criteria** — shown in a collapsible “Criteria” banner while you screen, so your rules stay in front of you.
- **Exclusion reasons** — a quick-select list. When you exclude an article you can *optionally* tag it with a reason (e.g. *Wrong Population*, *Wrong Study Design*), which feeds the exclusion breakdown and the PRISMA counts.

The abstract card is scrollable (rather than flick-to-swipe), so long abstracts are easy to read, and you decide with the **No / Maybe / Yes** buttons or the keyboard shortcuts. **Excluding never interrupts screening**: the exclude is saved instantly and the next abstract appears right away. If you defined exclusion reasons, a small panel appears below the card offering to tag the article you just excluded — click a reason or press the number keys 1–9 to record one, or simply keep screening (or press Esc / the × button) and no reason is stored. Tagging a reason is never required. If your very last decision is an exclude, the panel appears one final time — tag a reason or dismiss it, and the results screen opens.

## What Abstract Screening accepts as input

Abstract Screening works from a CSV, and it now supports **two** kinds of input.

**1. A reconciled CSV (from Synthesis).** If the CSV has a decision column — `final_decision` or `screening_decision` — SwipeScreen treats it as the output of a prior stage and loads **only the rows marked include or maybe**. Rows marked `exclude` (or anything else) are skipped. This is the standard “second pass on the survivors” workflow and is unchanged.

**2. A full, un-reconciled list (no decision column).** If the CSV has **no** `final_decision` and **no** `screening_decision` column, there is nothing to filter on — the list was never reconciled. In that case SwipeScreen screens **every row in the file**. This lets you run a single combined screening pass over an entire list from the start, without first going through Title Screening and Synthesis. Just upload a plain CSV (with **Title** and **Abstract** mapped) on the Abstract Screening path and start — every record becomes a card.

**Tip.** The deciding factor is only whether a decision column is present. A file with a `final_decision` / `screening_decision` column is always filtered to `include` / `maybe`; a file without one is always screened in full. If you have a reconciled file but want to re-screen every row, remove the decision column before uploading.

## The “nothing to screen” message

You will only ever see this blocking message:

*This CSV has a `final_decision` (or `screening_decision`) column, but no rows are marked `include` or `maybe`, so there is nothing to screen. Fix the decision values, or remove the column to screen every row in the file.*

in the one genuine case where the CSV **does** have a decision column but **none** of its rows are `include` or `maybe` (for example, every row is `exclude`). It does **not** appear for a valid file that simply has no decision column — that file is screened in full.

## Results and export

When a project is complete (or any time you open its Results screen) you get:

- **Decision counts** — Included, Maybe, and Excluded.
- **Exclusion reasons** — a ranked breakdown of the reasons you tagged.
- **Time stats** — total, average, and median time per card, plus the screening date range.

Two exports are available:

- **Download CSV** — your original columns plus screening columns: `screening_decision`, `exclusion_reason`, `reviewer`, `screening_phase`, `screening_timestamp`, and `time_on_card_seconds`. This is the file you feed into Synthesis (from Title Screening) or archive as your final record.
- **Save Progress (JSON)** — a full, resumable snapshot of the project, articles, and decisions. Reload it later with **Resume** on the home screen.

The tagged decisions and exclusion reasons map directly onto **PRISMA** flow counts (records screened, records excluded with reasons, and studies included; “maybe” is counted with the included group for a conservative screen).

## Privacy and data

- All data stays in your browser’s local IndexedDB storage.
- No account, no server, no upload.
- Because storage is local to one browser profile on one device, use **Save Progress (JSON)** to back up or move a project between machines, and keep your exported CSVs as the durable record.

## Quick reference

| I want to...                                 | Do this                                                                        |
|----------------------------------------------|--------------------------------------------------------------------------------|
| Screen titles fast                           | Title Screening → + New → upload CSV → swipe                                   |
| Combine two reviewers & measure kappa        | Synthesis → add both exported CSVs                                             |
| Second-pass the survivors by abstract        | Abstract Screening → + New → upload the reconciled CSV                         |
| Screen a whole list by abstract from scratch | Abstract Screening → + New → upload a plain CSV with <b>no</b> decision column |
| Resume an interrupted project                | Home → Resume → choose the saved JSON                                          |
| Get a PRISMA-ready record                    | Results → Download CSV                                                         |